

CS 6762
Fall 2023
SP, ML and FC
Homework 1

Topics covered: CPS hw/sw basics, sensors, basic sensing & sampling principles

Answer all questions. Work alone. Total 100 points. Type or write neatly.

1. What are some of the enabling HW/SW technologies that support the creation of the CPS/IOT revolution? (10 points)
2. What are the hardware components of a typical smart device? (be complete) (10 points)
3. What is energy scavenging? (5 points)
4. What are the main (14) properties of sensors that must be considered when choosing a sensor? (10 points)
5. What is the Nyquist sampling theorem? (5 points)
6. Describe aliasing and how you avoid it. (5 points)
7. Explain the concept of a virtual sensor? (5 points)
8. What are the typical parameters for an ADC? (5 points)
9. How does one connect peripherals/sensors to microcontrollers? (5 points)
10. Choose one of microphone, magnetometer, accelerometer or lidar. Search the web for descriptions of that class of sensor and describe the main issues related to that sensor type and how to program/use it. (15 points)
11. Assume you are part of a team for installing an IOT infrastructure for a smart city. What sensors would you include and why? (15 points)
12. Describe what sensors are needed and why for typical autonomous vehicles (driverless cars). (10 points)